

Capstone Two- Project Proposal

Problem Statement

A recent analysis documents a decrease in the county-level proportion of American kindergarteners that are up-to-date with the recommended measles-mumps-rubella (MMR) ([Dong et al. 2025](#)) since the COVID-19 pandemic. Analysis of national CDC vaccination coverage and exemption data for public school kindergarteners found small declines between [2019](#) and [2024](#) in the coverage of diphtheria, tetanus, and acellular pertussis vaccine (DTaP, from 94.9 to 92.5-0.4%), two doses of MMR (from 95.2% to 92.1%), as well as an increase in the proportion of kindergarten children with an exemption from one or more required vaccinations [Williams et al. 2025](#). Although there are variations across states and counties, these data suggest immunization rates among kindergarten aged children have not recovered to pre-pandemic levels ([article](#)) and fall short of the 95% threshold set by the WHO to prevent disease outbreaks ([WHO, UNICEF, 2025](#)).

In early 2025, a measles outbreak among children in Western Texas garnered national attention drawing attention to inadequate child immunization rates in American communities. Most of the identified measles cases have been unvaccinated or their vaccination status was unknown ([Texas Department of State Health Services](#), accessed Aug 8, 2025). As of August 2025, 762 cases have been confirmed, about 29.5% of them in children under the age of five years old. The pace of the outbreak in Texas, concentrated in Gaines County, has much decreased but there have since been outbreaks across the border in Lea County, New Mexico, in Kansas, and in Iowa ([Forbes article](#)).

This project proposal aims to investigate the factors that influence county-level child immunization rates against measles by triangulating available county-level data within the state of Texas.

Primary Questions:

1. What are the strongest predictors of high county-level measles vaccination rates across Texas being up-to-date on the full vaccine series?
2. If the relationships between predictors and county-level measles vaccination coverage have changed over the last five years, how have they changed?
3. What ability do logistic regression or random forest classifier models have in predicting county-level measles vaccination immunization coverage rates?
4. Are there "resilient" counties that do not fit the prevailing pattern and maintain high vaccination rates despite unfavorable factors? What county characteristics predict successful vaccination programs despite provider shortages?

Data Sources:

Dependent Variable:

The Texas Immunization Registry, ImmTrac2 is an opt-in registry that records the number of non-influenza vaccinations administered each month from 2019 to 2024. This includes the number of pediatric measles-containing vaccines (e.g. MMR) doses provided in each county to children 0-24 months old. Using this data source, my primary outcome of interest would be pediatric doses of MMR given to children 0-24 months old.

Another common source of immunization coverage data is the National Immunization Survey for children ([NIS-Child](#)) children aged 19 to 35 months. These data estimate immunization coverage rates for all childhood immunizations recommended by the Advisory Committee on Immunization Practices (ACIP) during a given year in each state and select territories of the United States. However they are not available by county. The NIS-Child survey's most recent available data is from their 2023 survey, so this limits the utility of that data source.

Independent Variables:

In terms of covariates or model features, there are several potential data sources that I could use to both validate NIS-C data and help explain variations in MMR coverage. It's also worth considering what other studies have used.

One study used both forward and backward step-wise selection of 105 potential covariates in order to estimate county-level coverage rates. The covariates they selected are marked with an asterisk (*) in the table below if I was able to locate similar data sources (Seeskin et al. 2025). In addition, they also selected the following indicators:

- Hepatitis A vaccine doses relative to the population of children
- Percentage of the civilian labor force employed in education/health care/social assistance
- Percentage of households with a female head

Type of county-level covariate	Covariate/Feature	Data Source
Health care access	Health insurance status * Number of uninsured children; Number of uninsured adults	countyhealthrankings.org
	Health Professional Shortage Area (HPSA) - Primary care physicians score;	Health Resources & Services Administration (HRSA) via BigQuery

	Medically Underserved Area (MUA) score - Primary care physicians	
Economic	Food environment index*; Food insecurity index; Limited access to health foods*	countyhealthrankings.org
	Median household income	countyhealthrankings.org
	Children in poverty	countyhealthrankings.org
Demographic	the percentage of population that is non-Hispanic, Black*, the percentage of the population that is non-Hispanic, White*;	countyhealthrankings.org
	Residential segregation (black/white)*	countyhealthrankings.org CDC
	Percentage of population that is children (0-18 years)	countyhealthrankings.org
	Percentage of residents in living in rural areas within the county	countyhealthrankings.org
Individual Psychosocial	COVID-19 vaccine hesitancy (from June 2021)	CDC
Other health outcomes	Infant mortality	countyhealthrankings.org
	Number of flu vaccinations administered	countyhealthrankings.org
	Number of infants with low birthweight	countyhealthrankings.org

Data wrangling

All data has been obtained in CSV format from respective websites. The ImmTrac2 data will be augmented with complementary sources such as the data on health care access, aggregate COVID-19 vaccine hesitancy rates, as well as indicators related to other socio-demographics, wealth, and health outcomes of Texan counties. Data sources which provide multiple records

(by FIPS code) per county will be collapsed and then all data files will be merged using a unique county index.

Exploratory analysis will involve looking at a merged and enriched data set and visualizing trends for anti-measles antigen containing vaccine coverage across several years (e.g. 2020-2025). EDA will then use scatter plots to visually examine the associations between measles immunization coverage rates and the covariates of interest in the table above.

Pre-processing and training data development

This step will involve building a pipeline that specifies imputation and feature selection parameters for logistic regression or random forest regressions models while performing crossover validation of longitudinal training data.

Modeling

In this step, I will fit the final model on all available and compare model fit or error statistics to those produced in previous model training.

Documentation

I will finish by creating a short report or slide deck on findings.